# Testing pure components

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

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TEACHER GUIDE · 1 OF 9

## Prepare the purpose, prerequisites and materials

> Prepare Testing pure components: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of testing pure components

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

### Least privilege and tool scopes

Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of least privilege and tool scopes

Reference example, not the only acceptable wording: Teacher explanation: Least privilege grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

### Configuration and secrets

Without opening the old lesson, explain one key idea from Configuration and secrets. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of configuration and secrets

Reference example, not the only acceptable wording: Teacher explanation: Configuration supplies non-secret settings; secrets provide privileged values. Keep secrets out of source, logs and client responses.

The toy service reads DATA_VERSION from configuration and a signing secret from server-only **input**. Validate required settings at startup and fail clearly if absent; do not substitute a published default secret.

### Requirements and service boundaries

Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of requirements and service boundaries

Reference example, not the only acceptable wording: Teacher explanation: A service boundary states who can request what, what data may be read and which actions are outside scope. Requirements must be observable.

The fictional student timetable service reads a public schedule and returns a meeting time or review status. It cannot edit schedules or email users. Test success and refusal before adding model behaviour.

Two 45-minute sessions. Pure components return results from supplied **input**s without external side effects. They are easier to test with exact expected values.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

### Outcomes

Explain the **mechanism** and terminology of testing pure components

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

**Prerequisite answer**

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

**End goal for the pictorial case**

Explain testing pure components using the deciding step in this fictional service.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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## Explain and model the mechanism

> Model Testing pure components using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Pure components return results from supplied **input**s without external side effects. They are easier to test with exact expected values.

### Explain Testing pure components · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Define remaining(capacity, used)=capacity-used for integer **input**s satisfying 0<=used<=capacity. Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries.

## Explain Testing pure components · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

| Term | Meaning |
| --- | --- |
| Pure function | A computation with no externally observable side effects |
| Boundary case | An input at a contract edge |
| Expected outcome | The result or error specified before execution |

### Testing pure components — evidence trace

**INPUT AND SCOPE**

Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13.

**CHECK OR CALCULATION**

Define remaining(capacity, used)=capacity-used for integer inputs satisfying 0<=used<=capacity. Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries.

**CONCLUSION AND LIMIT**

Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

## Explain Testing pure components · 3

Guided: Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

Qualification: Passing unit cases for a **pure function** does not establish integration permissions or model-dependent behaviour.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13.

Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

## Model the reasoning aloud · 4

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

## Model the reasoning aloud · 5

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def nonnegative_integer(value):
    return type(value) is int and value >= 0

def remaining(capacity, used):
    if not nonnegative_integer(capacity) or not nonnegative_integer(used) or used >
capacity:
        raise ValueError('capacity/used')
```

```
    return capacity - used

print(remaining(12, 5))
```

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Testing pure components

Test a room formatter alone. A **pure function** returns "Room " plus a supplied positive integer.

### The facts we will use

> #### Supplied facts
>
> Accepted **input**s are positive integers. Expected **output**s: 7→"Room 7"; 12→"Room 12".

AI ACADEMY / AI-6/7

# Testing pure components

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.

### 1. Call with 7

The prefix and supplied numeral produce Room 7.

### 2. Compare exact text

The expected text includes one space before 7.

### 3. Call with 12

The same fixed procedure produces Room 12.

**CHANGE ONE THING**

Supply zero instead of 7.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Call with 7 | The prefix and supplied numeral produce Room 7. |
| 2. Compare exact text | The expected text includes one space before 7. |
| 3. Call with 12 | The same fixed procedure produces Room 12. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

A pure component depends on its accepted **input**, not a clock, network or model. Isolating it makes failures easier to locate.

### What this example does not establish

#### Keep the stated limit

These tests cover only stated cases; they do not test the full AI service or every invalid value.

### Change one thing in the pictured case

#### Change one thing

Change: Supply zero instead of 7.

#### Predicted result

**Prediction**: Reject the **input** under the stated positive-integer contract.

#### Why the result follows

Reason: Zero has the right type but violates the value requirement.

### Ask while showing the picture

#### Ask while showing the picture · 6

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

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## Demonstrate and guide practice

> Demonstrate Testing pure components, reduce support deliberately and check individual reasoning before discussing answers.

| Time or stage | Teacher action |
| --- | --- |
| Session 1: 0–7 minutes | Retrieve the prerequisite; identify the input, procedure and evidence source. |
| 7–18 minutes | Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation. |
| 18–32 minutes | Trace the declared procedure and worked case. Complete W2, recording intermediate steps. |
| 32–45 minutes | Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations. |
| Session 2: 0–8 minutes | Retrieve the worked reasoning and collect the guided response before discussing it. |
| 8–20 minutes | Compare a proposed boundary variant with the original case; identify what remains fixed. |
| 20–35 minutes | Collect W4 independently. Require the procedure trace and the bounded conclusion. |
| 35–45 minutes | Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts. |

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

### Reduce help deliberately

#### Completion practice

Completion task: Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13. Show the intermediate steps and give the conclusion with its scope.

### Reasoning to listen for

Teacher check: Teacher explanation: Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

#### Individual reasoning check

Use the worked case for Testing pure components. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Expected repair

Expected correction: Passing unit cases for a **pure function** does not establish integration permissions or model-dependent behaviour.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

#### Pictorial practice question

Predict the formatter **output**s for 7 and 12, including spacing. Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?

**Reasoned teacher answer**

The prefix and supplied numeral produce Room 7. The expected text includes one space before 7. The same fixed procedure produces Room 12. Changed case: Reject the **input** under the stated positive-integer contract.

Zero has the right type but violates the value requirement.

Changed-case reasoning: Reject the **input** under the stated positive-integer contract. Zero has the right type but violates the value requirement.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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## Run the investigation and diagnose misconceptions

> Run the Testing pure components investigation and use the first unsupported step to choose a teaching response.

The worked result establishes every future case: Passing unit cases for a **pure function** does not establish integration permissions or model-dependent behaviour.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

### Match the support to the first error

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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## Assess and explain the answers

> Use the Testing pure components questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Guided and exit questions

Guided: Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

Qualification: Passing unit cases for a **pure function** does not establish integration permissions or model-dependent behaviour.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

### W1 Explain the mechanism

#### Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

#### Teacher reasoning · W1

Teacher explanation: Pure components return results from supplied **input**s without external side effects. They are easier to test with exact expected values.

Define remaining(capacity, used)=capacity-used for integer **input**s satisfying 0<=used<=capacity. Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries.

### W2 Trace the worked case

#### Workbook question · W2

Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13. Show the intermediate steps and give the conclusion with its scope.

#### Teacher reasoning · W2

Teacher explanation: Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

### W3 Audit the limit

#### Workbook question · W3

A report omits this qualification: Passing unit cases for a **pure function** does not establish integration permissions or model-dependent behaviour.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

#### Teacher reasoning · W3

Teacher explanation: Passing unit cases for a **pure function** does not establish integration permissions or model-dependent behaviour.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

### W4 Apply a new case independently

#### Workbook question · W4

Capacity 9 with used 0, 9, 10 and Boolean True. Give outcomes.

#### Teacher reasoning · W4

Teacher explanation: 9, 0, range error, type error. Pure arithmetic must not silently admit bool as a count.

### W5 Design a revealing follow-up

#### Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its **expected outcome** before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

#### Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Passing unit cases for a **pure function** does not establish integration permissions or model-dependent behaviour.

The new case should test this particular **mechanism** rather than merely restate the original result.

### Independent assessment and fresh retry

#### Teacher key · Independent assessment and fresh retry

W4: 9, 0, range error, type error. Pure arithmetic must not silently admit bool as a count. Require intermediate reasoning and the stated scope. Fresh retry: Capacity 4, used 2; capacity -1, used 0.

Expected reasoning: 2; invalid negative capacity. No storage or network action is needed. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

### Assess independent transfer

#### Independent question · withhold the key

Selected case: Capacity 9 with used 0, 9, 10 and Boolean True. Give outcomes.

### Selected case answer

#### Teacher key · Selected case answer

Teacher explanation: 9, 0, range error, type error. Pure arithmetic must not silently admit bool as a count.

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

#### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect the project, support and next lesson

> Connect Testing pure components to Grounded Learning Service, adapt support and plan a later changed-case check.

Add a labelled testing pure components **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of testing pure components.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Testing model-dependent behaviour. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

### Plan the delayed return

Suggested return points in this level: ai-6/8, ai-6/10, ai-6/13. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Testing pure components lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Explain the **mechanism** and terminology of testing pure components

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13. Show the intermediate steps and give the conclusion with its scope.

### Supported practice answer

Reasoned practice answer: Teacher explanation: Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

Check every learner before discussion: Use the worked case for Testing pure components. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: Passing unit cases for a **pure function** does not establish integration permissions or model-dependent behaviour.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: Supply zero instead of 7. Require a **prediction** and a reason before comparing. Expected result: Reject the **input** under the stated positive-integer contract.

Reason: Zero has the right type but violates the value requirement.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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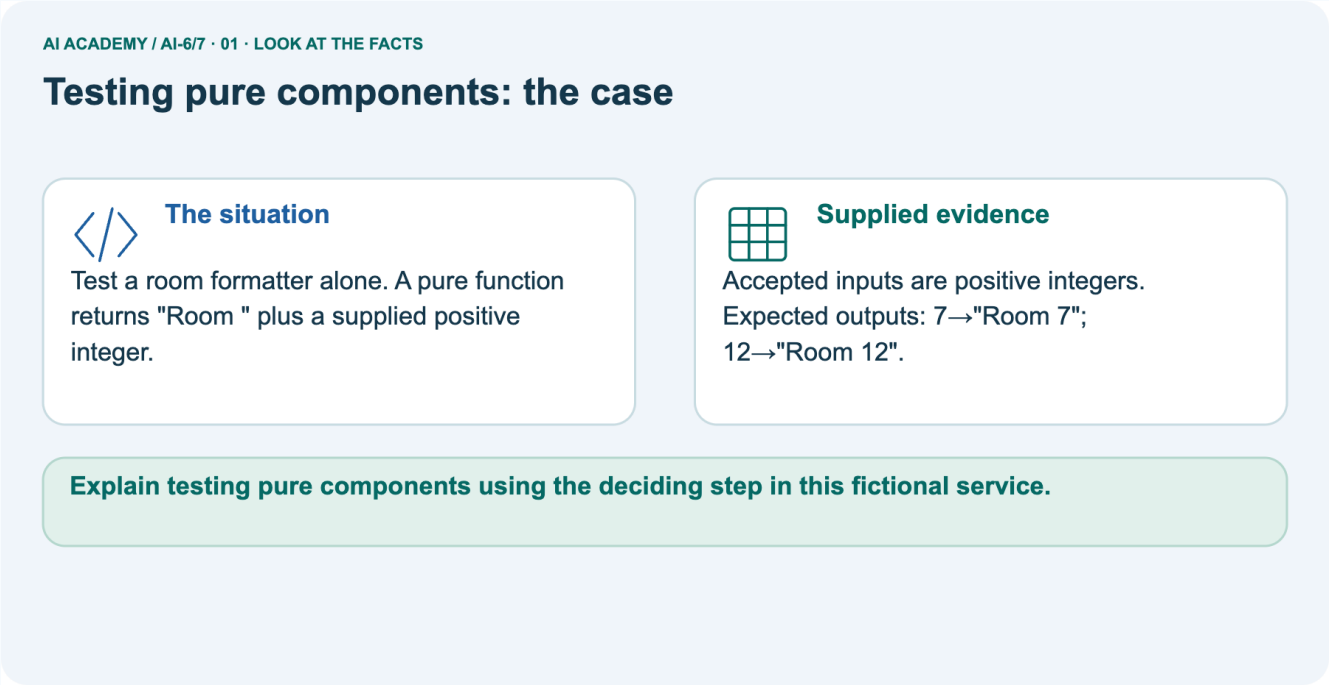


## Visual concept journey

> Show one Testing pure components graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

## 01. Testing pure components: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

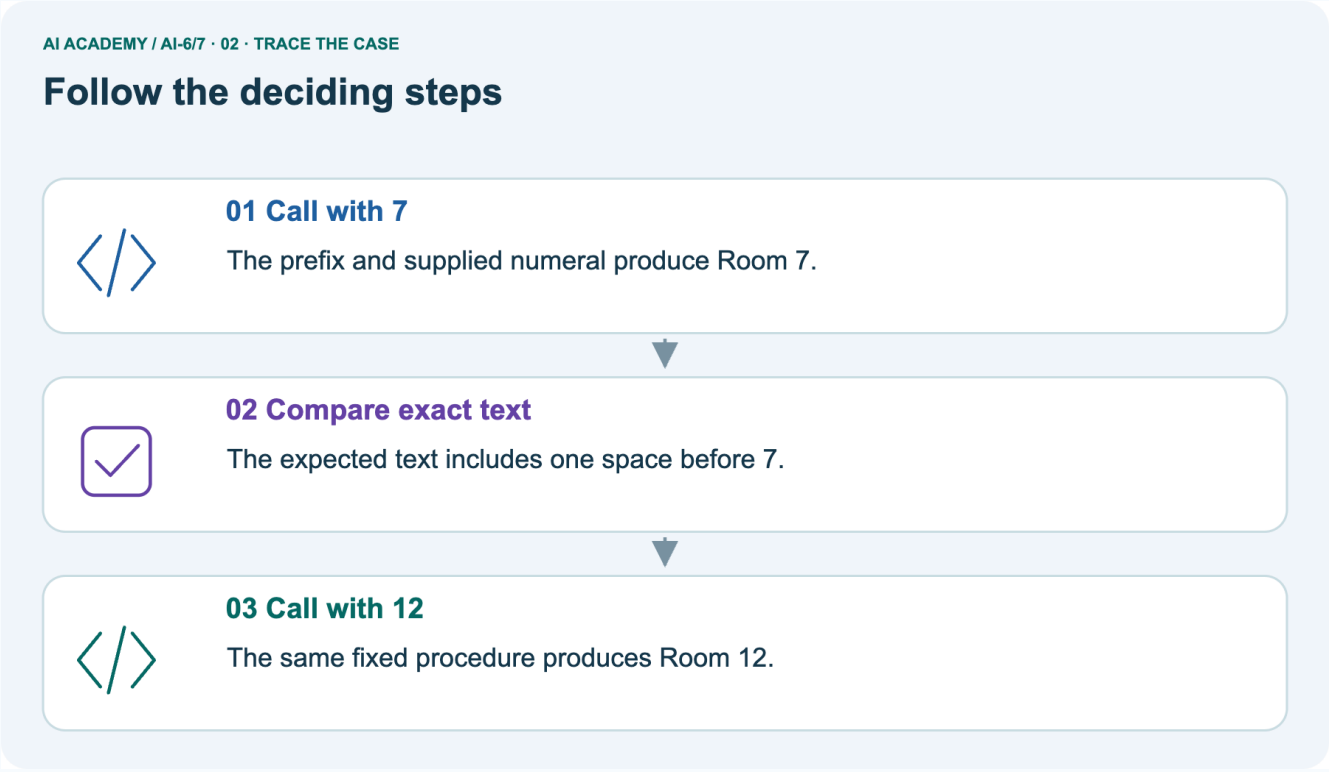
Test a room formatter alone. A **pure function** returns "Room " plus a supplied positive integer. Supplied facts: Accepted **input**s are positive integers. Expected **output**s: 7→"Room 7"; 12→"Room 12".

Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

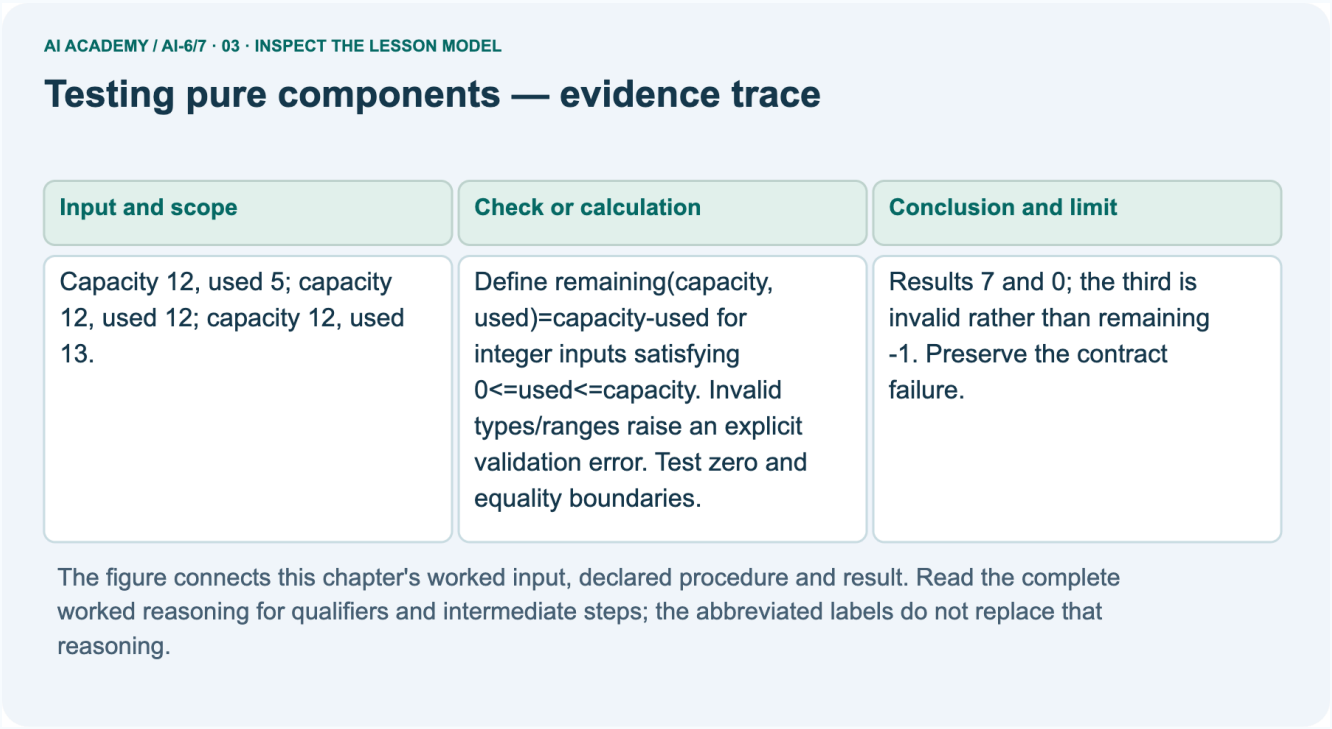
Call with 7: The prefix and supplied numeral produce Room 7. → Compare exact text: The expected text includes one space before 7. → Call with 12: The same fixed procedure produces Room 12.

Point to each numbered stage and explain which recorded **input** it uses.

#### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 03. Testing pure components — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

Testing pure components — **evidence** trace. Capacity 12, used 5; capacity 12, used 12; capacity 12, used 13. | Define remaining(capacity, used)=capacity-used for integer **input**s satisfying 0<=used<=capacity.

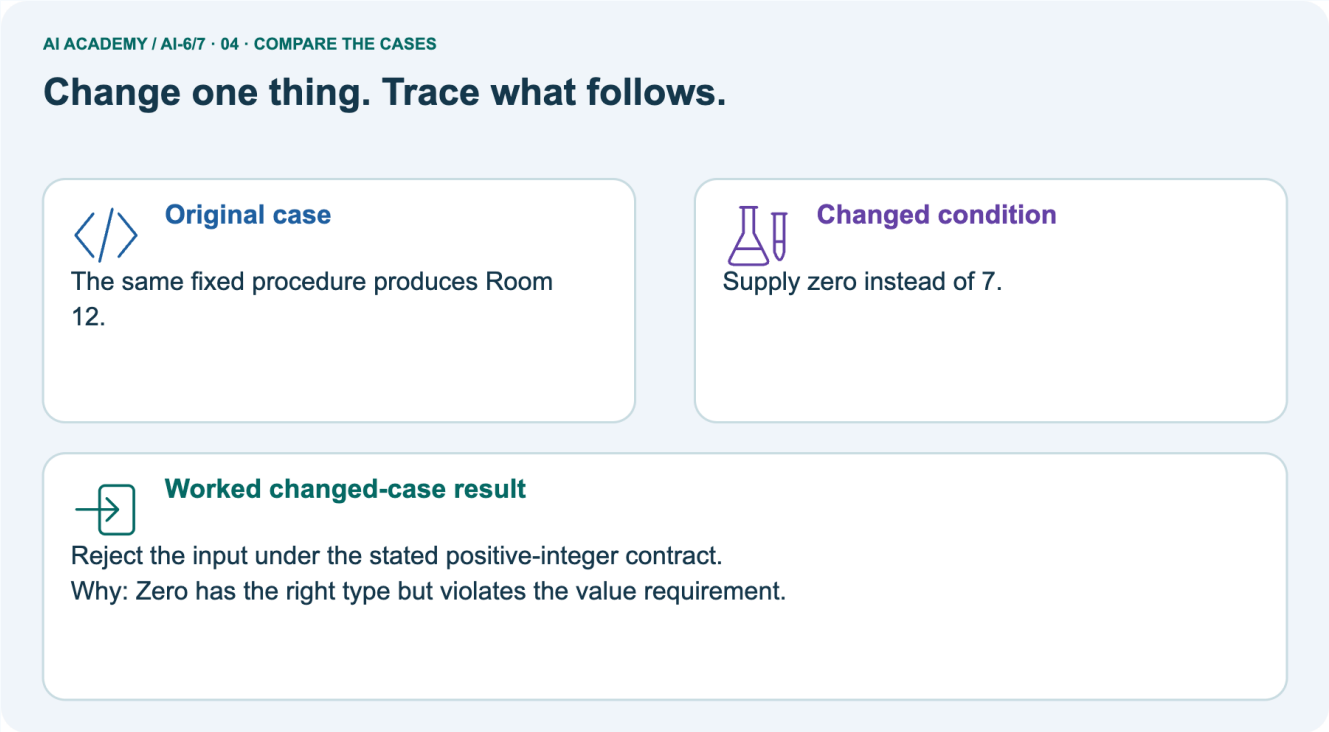
Invalid types/ranges raise an explicit validation error. Test zero and equality boundaries. | Results 7 and 0; the third is invalid rather than remaining -1. Preserve the contract failure.

> Read the labels and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The same fixed procedure produces Room 12. Changed condition: Supply zero instead of 7. Expected result: Reject the **input** under the stated positive-integer contract.

Reason: Zero has the right type but violates the value requirement.

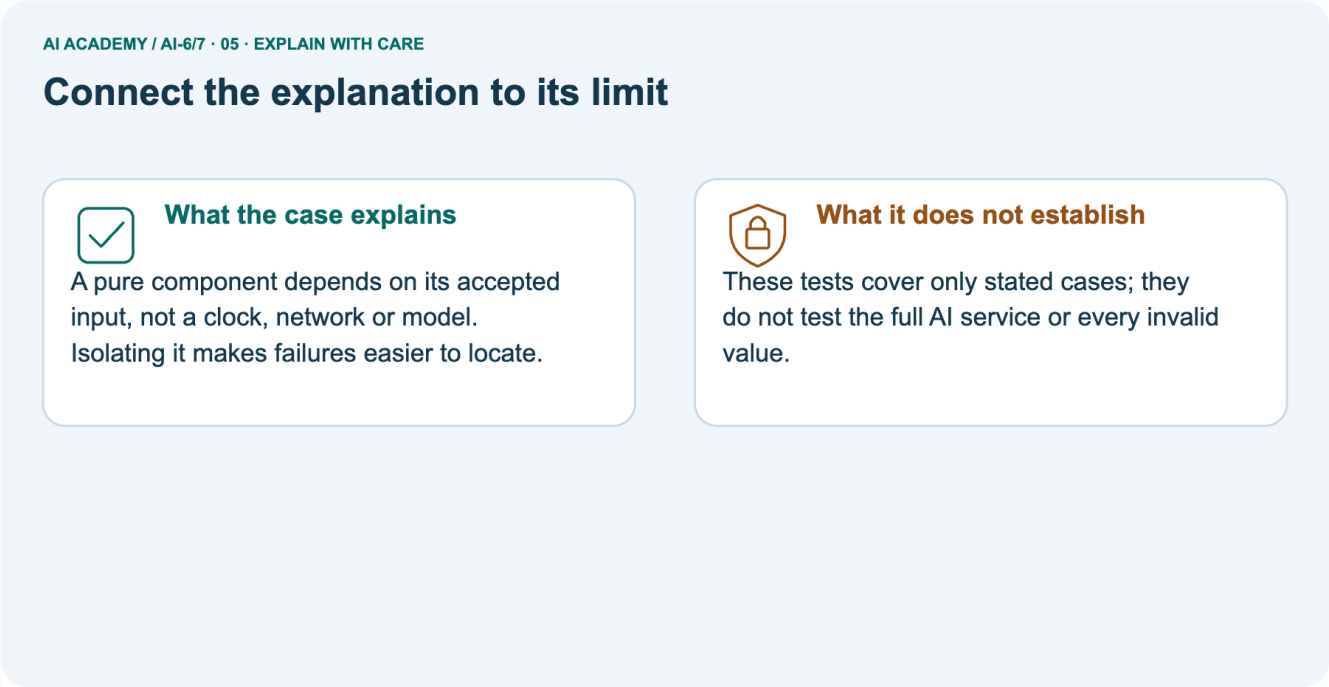
> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Zero has the right type but violates the value requirement.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

A pure component depends on its accepted **input**, not a clock, network or model. Isolating it makes failures easier to locate.

Limit: These tests cover only stated cases; they do not test the full AI service or every invalid value.

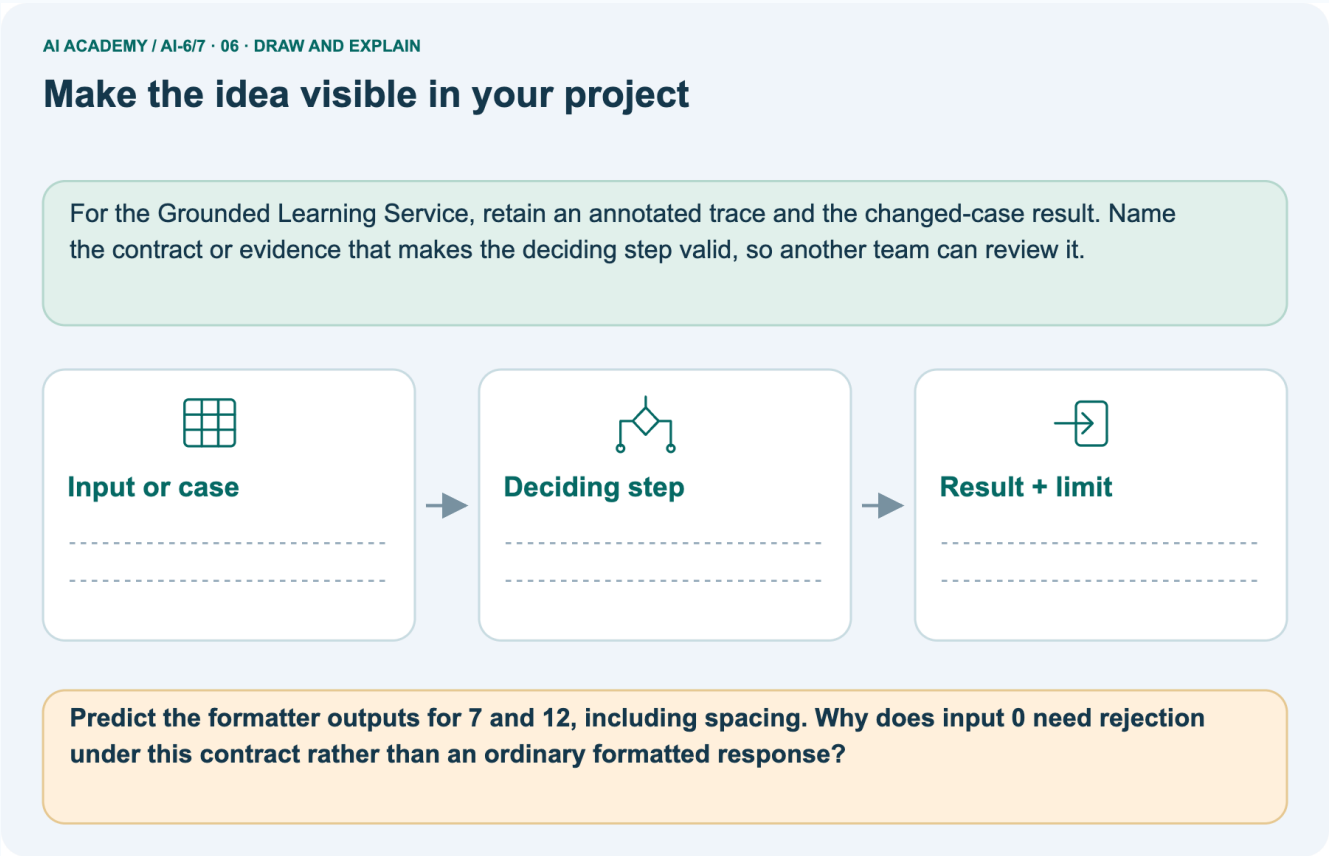
Explain the case, then point to a limit on the claim.

#### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Keep the stated scope: These tests cover only stated cases; they do not test the full AI service or every invalid value.

### 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Predict the formatter **output**s for 7 and 12, including spacing.

Why does **input** 0 need rejection under this contract rather than an ordinary formatted response?

> Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The prefix and supplied numeral produce Room 7. The expected text includes one space before 7. The same fixed procedure produces Room 12.

Changed case: Reject the **input** under the stated positive-integer contract. Zero has the right type but violates the value requirement.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.